Report 2: UML Class Diagram Report

Hospital Management System (HMS) — Class Structure & Relationships

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1. Class Diagram Specifications

Patient

```
- patientId: String
- fullName: String
- age: int
- gender: String
- phone: String
```

```
+ register(): void
+ updateProfile(): void
+ viewHistory(): List
```

Doctor

```
- doctorId: String
- fullName: String
- specialization: String
- email: String
```

```
+ checkAvailability(): bool
+ addRecord(): void
```

Appointment

```
- appointmentId: String
- appointmentDate: Date
- status: String
```

```
+ schedule(): void
+ cancel(): void
+ reschedule(): void
```

MedicalRecord

```
- recordId: String | - diagnosis: String
- prescription: String | - recordDate: Date
```

```
+ updateRecord(): void | + printRecord(): String
```

Billing

```
- billId: String | - amount: double
- issueDate: Date | - isPaid: boolean
```

```
+ generateInvoice(): void | + processPayment(): void
```

2. Class Relationships & Multiplicities

Source Class	Relationship Type	Target Class	Multiplicity	Description
Patient	Association	**Appointment**	1 to 0..*	A patient can schedule zero or many appointments; each appointment belongs to 1 patient.
Doctor	Association	**Appointment**	1 to 0..*	A doctor manages zero or many scheduled appointments; each appointment has 1 doctor.
Patient	Composition	**MedicalRecord**	1 to 0..*	A patient owns medical records. If the patient record is deleted, their medical records are removed.
Appointment	Association / Aggregation	**Billing**	1 to 0..1	Each completed appointment generates exactly one bill statement.

3. Relationship Rationale

- **Patient — Appointment (1 : 0..*):** Enables patients to book multiple consultations over time while retaining individual appointment accountability.
- **Doctor — Appointment (1 : 0..*):** Guarantees that each consultation is directed by a specific physician without schedule overlapping.
- **Patient — MedicalRecord (1 : 0..* - Composition):** Strong ownership relationship where medical records cannot exist independently of a registered patient.
- **Appointment — Billing (1 : 0..1):** Ensures automated billing creation immediately following a patient appointment session.